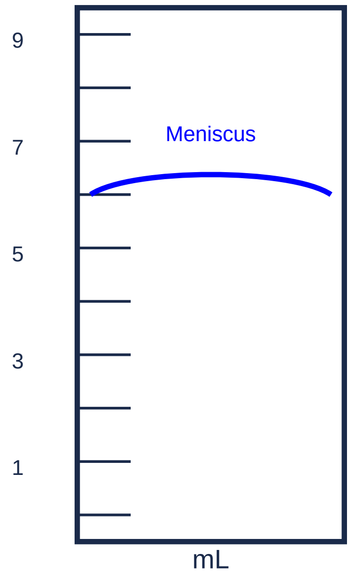
Unit 2: Measurements and Scientific Methods

Complete Study Notes – Grade 9 Chemistry

Measuring Cylinder



2.1 Measurements and Units in Chemistry

Measurement is the comparison of a physical quantity to be measured with a unit of measurement – a fixed standard. In science, units are essential to state measurements correctly. A number without a unit is meaningless.

In traditional markets, people buy and sell goods by estimating their size using indigenous methods. In chemistry, however, we use standardised instruments for precise measurements.

Common measuring devices found in a chemistry laboratory:

- **Meter stick –** measures length.
- **Burette, pipette, graduated cylinder, and volumetric flask –** measure volume.
- **Balance –** measures mass.
- **Thermometer –** measures temperature.

The instruments above provide measurements of **macroscopic properties**, which can be determined directly. **Microscopic properties**, on the atomic or molecular scale, must be determined by indirect methods.

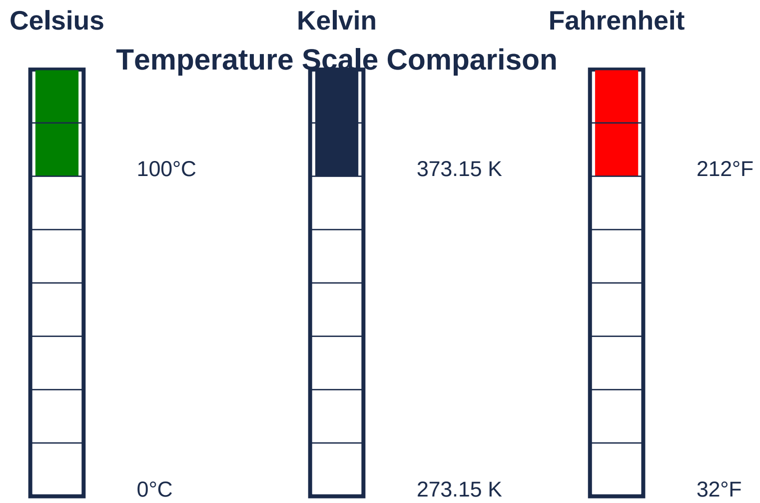
2.1.1 SI Units (The International System of Units)

In 1960, the General Conference of Weights and Measures proposed the International System of Units (SI). It is based on metric units and uses decimal relationships (powers of 10).

Base Quantity	Name of Unit	Symbol
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of substance	Mole	mol
Luminous intensity	Candela	cd

Heat and Temperature

Temperature measures the intensity of heat – the 'hotness' or 'coldness' of a body. **Heat** is a form of energy that always flows spontaneously from a hotter body to a colder body – never in the reverse direction.



Between the freezing point of water (0°C) and the boiling point of water (100°C), there are 100 steps on the Celsius scale and 100 kelvins on the Kelvin scale. Thus, the 'degree' is the same size on both scales. But every Kelvin temperature is 273.15 units above the corresponding Celsius temperature.

Conversions:

K = °C + 273.15 and **°C = K – 273.15**

Comparing Fahrenheit and Celsius: there are 180 Fahrenheit degrees between the freezing and boiling points of water, and 100 Celsius degrees. Therefore, 1°F = 5/9°C and 1°C = 9/5°F.

°F = (°C × 9/5) + 32 and **°C = (°F – 32) × 5/9**

Example: When the temperature reaches '100.0 °F in the shade', it's hot. What is this temperature on the Celsius scale?

Solution: °C = (100.0 – 32) × 5/9 = 68 × 5/9 = 37.8 °C

2.1.2 Derived Units

Derived units are obtained by combining SI base units using mathematical equations. Once base units have been defined, you can derive other units by using the base units in equations that define other physical quantities.

For example, area is defined as length times width. Therefore, the SI unit of area is **m × m = m²**. Speed is distance divided by time, so the SI unit of speed is **m/s**.

Quantity	Definition	SI Unit
Area	Length × Width	m^2
Volume	Length × Width × Height	m^3
Density	Mass / Volume	kg/m^3
Speed	Distance / Time	m/s
Acceleration	Speed change / Time	m/s^2
Force	Mass × Acceleration	$kg{\cdot}m/s^2$ (N)
Pressure	Force / Area	$kg/(m{\cdot}s^2)$ (Pa)

Energy	Force × Distance	kg·m²/s² (J)

Volume:

Volume is defined as length cubed and has the SI unit of cubic meter (m^3). This is too large for normal laboratory work, so we use cubic decimeters (dm^3) or cubic centimetres (cm^3). Traditionally, chemists use the **litre (L)**, which is equal to a cubic decimetre.

1 L = 1 dm^3 = 1000 cm^3 and **1 mL = 1 cm^3**

Density:

The **density** of an object is its mass per unit volume:

d = m / V

Density is an important characteristic property of a material. Water has a density of 1.000 g/cm^3 at 4°C. Lead has a density of 11.3 g/cm^3 at 20°C. Density can be used to identify a substance, determine purity, and distinguish between materials.

Example: A colorless liquid, believed to be one of several substances, was found to have a mass of 30.5 g and a volume of 35.1 mL. What is its density?

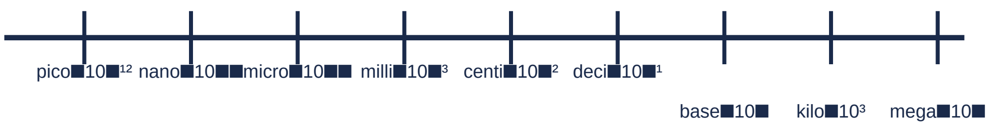
Solution: d = 30.5 g / 35.1 mL = 0.869 g/mL – this matches the density of toluene.

Exercise: A piece of metal wire has a volume of 20.2 cm^3 and a mass of 159 g. The metal is manganese (7.21 g/cm^3), iron (7.87 g/cm^3), or nickel (8.90 g/cm^3). From which metal is the wire made?

Solution: d = 159 g / 20.2 cm^3 = 7.87 g/cm^3 – the wire is made of **iron**.

2.1.3 Common Prefixes Used in SI Units

SI Prefixes (decreasing → ← increasing)



Factor	Prefix	Symbol	Example
10^{12}	Tera	T	1 Tm = 1×10^{12} m
10■	Giga	G	1 Gm = 1×10■ m
10■	Mega	M	1 Mm = 1×10■ m
10^3	Kilo	k	1 km = 1×10^3 m
10^2	Hecto	h	1 hm = 1×10^2 m
10^1	Deca	da	1 dam = 1×10^1 m

$10^{\blacksquare 1}$	Deci	d	1 dm = $1\times10^{\blacksquare 1}$ m
$10^{\blacksquare 2}$	Centi	c	1 cm = $1\times10^{\blacksquare 2}$ m
$10^{\blacksquare 3}$	Milli	m	1 mm = $1\times10^{\blacksquare 3}$ m
$10^{\blacksquare\blacksquare}$	Micro	µ	1 µm = $1\times10^{\blacksquare\blacksquare}$ m
$10^{\blacksquare\blacksquare}$	Nano	n	1 nm = $1\times10^{\blacksquare\blacksquare}$ m
$10^{\blacksquare 12}$	Pico	p	1 pm = $1\times10^{\blacksquare 12}$ m

Examples of SI prefixes:

- 1 Megawatt = 1,000,000 watts
- 1 kilogram = 1,000 grams
- 1 micrometre (µm) = 1/1,000,000 metre

2.1.4 Uncertainty in Measurements

Whenever you measure something, there is always some uncertainty. There are two categories of uncertainty:

1. Systematic uncertainties – consistently cause the value to be too large or too small. Examples include reaction time, inaccurate meter sticks, optical parallax, and miscalibrated balances. In principle, systematic uncertainties can be eliminated if you know they exist.

2. Random uncertainties – variations in measurements that occur without a predictable pattern. They arise from the estimated part of the measurement. Random uncertainty can be reduced but never eliminated.

The uncertainty is expressed as an interval within which the result can be found with a given probability. The result will be within the interval, but all values within the interval have the same probability to represent the result.

Example – Volume measurement:

Using a graduated cylinder with a scale that gives an uncertainty of 1 mL, a volume of 6 mL is expressed as (6 ± 1) mL. Using a more precise cylinder with finer divisions, the volume might be (6.0 ± 0.1) mL.

Calculating uncertainty:

Example: A student measures the diameter of a metal canister using a ruler graduated in mm and records: 66 mm, 65 mm, 61 mm. The mean is 64 mm. The uncertainty can be calculated as:

- **Half range:** (66 – 61) / 2 = 2.5 mm → 64 ± 2.5 mm
- **Reading furthest from mean:** 64 – 61 = 3 mm → 64 ± 3 mm
- **Resolution of instrument:** ruler reads to 0.5 mm → 64 ± 0.5 mm

Percent uncertainty:

Percent uncertainty = (absolute uncertainty / measurement) × 100

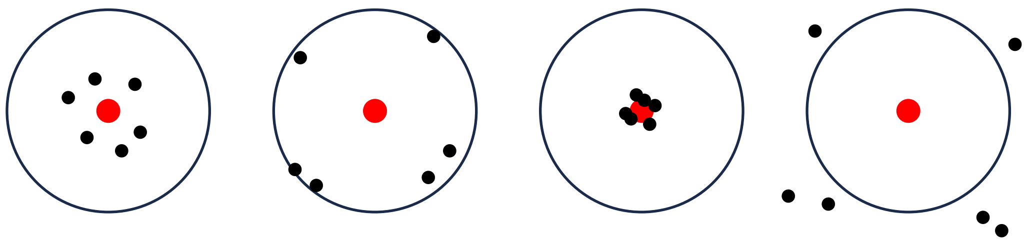
Example: A mass of 23.25 g weighed on a balance with uncertainty 0.05 g: % uncertainty = (0.05 / 23.25) × 100 = 0.2%

2.1.5 Precision and Accuracy

Accuracy refers to the closeness of a measured value to the true value.

Precision refers to the closeness of a set of values obtained from identical measurements of a quantity – the reproducibility of results.

The goal of scientists is to obtain measured values that are **both accurate and precise**.



a) Precise but Inaccurate b) Accurate but Imprecise c) Both Precise & Accurate d) Neither Precise nor Accurate

Example: A 2-carat diamond has a true mass of 400.0 mg. A jeweler repeatedly weighed it and obtained 450.0 mg, 459.0 mg, and 463.0 mg. These measurements are precise (close to each other) but not accurate (far from the true value).

Example: Three analyses of a copper penny gave 93.2%, 92.9%, and 93.5% zinc, while the true composition was 97.6% zinc. The results are precise (close to each other) but not accurate (far from the true value).

2.1.6 Significant Figures

Significant figures are the meaningful digits in a measured or calculated quantity. When significant figures are used, the last digit is understood to be uncertain.

Rules for determining significant figures:

1. **Any non-zero digit is significant.** (845 cm → 3 sig fig; 1.234 kg → 4 sig fig)
2. **Zeros between non-zero digits are significant.** (606 m → 3 sig fig; 40,501 kg → 5 sig fig)
3. **Zeros to the left of the first non-zero digit are not significant.** They indicate the placement of the decimal point. (0.08 L → 1 sig fig; 0.0000349 g → 3 sig fig)
4. **If a number is greater than 1, all zeros written to the right of the decimal point count as significant.** (2.0 mg → 2 sig fig; 40.062 mL → 5 sig fig; 3.040 dm → 4 sig fig)
5. **For numbers that do not contain decimal points, trailing zeros may or may not be significant.** 400 cm could have 1, 2, or 3 sig fig – use scientific notation to avoid ambiguity.

Rules for calculations with significant figures:

Addition and subtraction: The answer cannot have more digits to the right of the decimal point than either of the original numbers.

Example: 89.332 + 1.1 = 90.432 → rounded to 90.4 (one decimal place)

Example: 2.097 – 0.12 = 1.977 → rounded to 1.98 (two decimal places)

Multiplication and division: The number of significant figures in the final product or quotient is determined by the original number that has the smallest number of significant figures.

Example: $2.8 \times 4.5039 = 12.61092 \rightarrow$ rounded to 13 (2 sig fig)

Example: $6.85 \div 112.04 = 0.0611388789 \rightarrow$ rounded to 0.0611 (3 sig fig)

Example: Determining significant figures

- 394 cm → 3 sig fig
- 5.03 g → 3 sig fig
- 0.714 m → 3 sig fig
- 0.052 kg → 2 sig fig
- 2.720×10^{22} atoms → 4 sig fig
- 3000 mL → ambiguous – use scientific notation

2.1.7 Scientific Notation and Decimal Places

Scientific notation is used to express very large or very small numbers conveniently. The format is: **M × 10■**, where M is a number between 1 and 10, and n is an integer.

Example: 197 grams of gold contains approximately 602,000,000,000,000,000,000,000 gold atoms. In scientific notation: **6.02×10^{23}**.

The mass of one gold atom is approximately 0.000000000000000000000327 gram. In scientific notation: **$3.27 \times 10^{■22}$**.

Example: Unit conversion

The Angstrom (Å) is a unit of length, 1 Å = $1\times10^{■1■}$ m. The radius of a phosphorus atom is 1.10 Å. What is this distance in centimetres and nanometres?

Solution: 1.10 Å × ($1\times10^{■1■}$ m / 1 Å) × (1 cm / $1\times10^{■2}$ m) = **$1.10\times10^{■■}$ cm**

1.10 Å × ($1\times10^{■1■}$ m / 1 Å) × (1 nm / $1\times10^{■■}$ m) = **$1.10\times10^{■1}$ nm = 0.110 nm**

2.2 Chemistry as Experimental Science

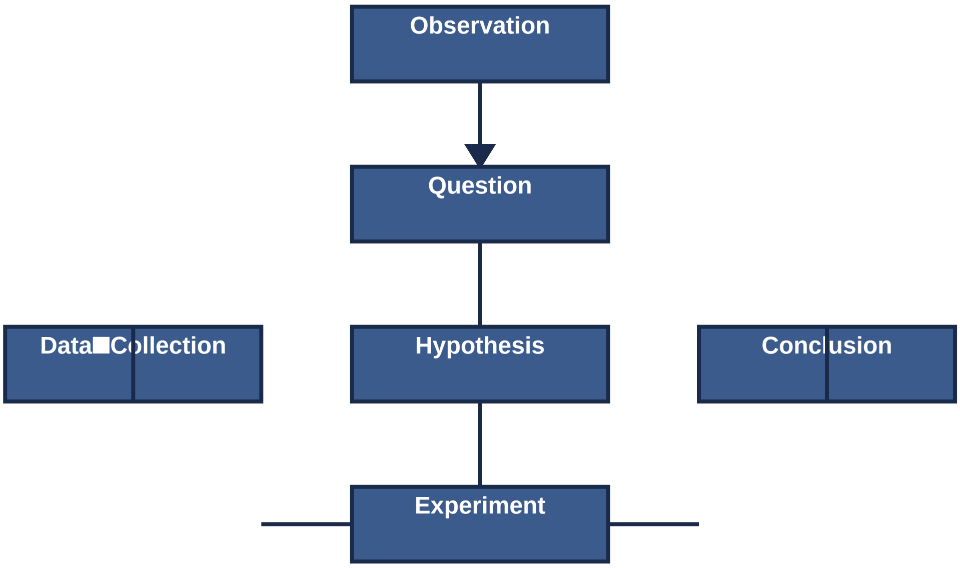
Chemistry is largely an **experimental science**; a great deal of knowledge comes from laboratory research. Today's chemists also use computers to study microscopic structure and sophisticated electronic equipment to analyse pollutants.

Chemists participate in the development of new drugs, agricultural research, and environmental solutions. Because of its diverse applications, chemistry is often called the '**central science**'.

2.2.1 The Scientific Method

The **scientific method** is a process by which scientists investigate, verify, or construct an accurate and reliable version of natural phenomena.

The Scientific Method



The four main steps of the scientific method:

1. Observation and formulation of a Question: Identify an observable aspect of the universe and ask a question about it. Example: 'Why is the sky blue?' or 'Why does air become invisible?'

2. Data Collection and Hypothesis: Collect all related data and formulate a hypothesis – a testable explanation or educated guess based on the observation. The hypothesis could be the cause of the phenomenon, its effect, or its relation to any other phenomenon.

3. Testing the Hypothesis: Conduct experiments to determine whether the hypothesis agrees with or contradicts observations in the real world. The confidence in the hypothesis increases or decreases based on the results.

4. Analysis and Conclusion: Use mathematical and scientific procedures to determine the results. If the data is consistent with the hypothesis, it is accepted. If not, it is rejected or modified and analyzed again. A hypothesis cannot be proved or disproved by a single experiment; repeated experiments with no discrepancies lead to a **theory**.

Example – Scientific method in everyday life:

A student wants to test if temperature affects seed germination. They observe that seeds planted in different locations germinate at different times. They hypothesise: 'Seeds germinate faster at room temperature than in the fridge or kitchen.' They experiment by planting seeds in three different temperature conditions. They collect data (time and length of germination), draw conclusions, and communicate the findings.

2.2.2 Some Experimental Skills in Chemistry

Laboratory Safety Rules

The chemical laboratory can be no more dangerous than any other classroom if safety rules are always observed. Most are based on simple common sense.

1. **Responsible behaviour** – essential. Spilled acids and broken glass from thoughtless actions are too dangerous.
2. **Wear approved eye protection** at all times in the laboratory and in areas where chemicals are stored or handled.
3. **Perform no unauthorized experiments** – use only the quantities instructed.
4. **Do not smoke** – it is a fire hazard and draws chemicals into the lungs.
5. **In case of fire or accident, call the teacher at once** – know the location of safety equipment.
6. **Report all injuries** at once. Get medical treatment for cuts, burns, or fume inhalation.
7. **Do not eat or drink** in the laboratory.
8. **Avoid breathing fumes** of any kind.
9. **Never use mouth suction** in filling pipets – always use a suction device.
10. **Never work alone** – there must be at least one other person present.
11. **Wear shoes** – bare feet and sandals are prohibited.
12. **Confine long hair and loose clothing** – they may catch fire or become contaminated.
13. **Keep your work area neat** – clean up spills and broken glass immediately.
14. **Be careful when heating liquids** – add boiling chips to avoid 'bumping'. Flammable liquids must not be heated over an open flame.
15. **Always pour acids into water** when mixing (acid into water, never water into acid).
16. **Do not force rubber stoppers** onto glass tubing – lubricate and protect your hands.
17. **Dispose of excess liquid reagents** by flushing small quantities down the sink. Dispose of solids in crocks. Never return reagents to bottles.
18. **Read the experiment carefully** before coming to the laboratory – an unprepared student is a hazard.
19. **Test tubes being heated should never be pointed at anyone.**
20. **If you have a cut on your hand, cover it** with a bandage or wear gloves.
21. **Finally, and most importantly, think about what you are doing.** Plan ahead.

Common Laboratory Equipment:

- **Beaker** – used for mixing, stirring, and heating chemicals. Has a spout for pouring.
- **Graduated cylinder** – used to measure volume (less precise than a pipet or burette).
- **Volumetric flask** – used to prepare precise volumes of solutions.
- **Burette** – used for titrations to deliver precise volumes.
- **Pipette** – used to transfer exact volumes of liquid.
- **Watch glass** – used to cover beakers to prevent contamination or splashing.

2.2.3 Writing a Laboratory Report

The Pre-laboratory Report:

The prelab report is completed before the experiment begins. Its purpose is to ensure familiarity with the procedure and provide for more efficient use of laboratory time. Prelab questions are answered after careful reading of the introduction and procedure. Sample calculations may be included to provide awareness of data that needs to be collected and how it is treated.

The Laboratory Report:

- A laboratory report is a final draft – written in ink or typed. No erasures or crossed-out areas.
- Keep an initial draft in your laboratory notebook – you may need to rewrite it.
- All data should be presented with the correct significant figures and units.
- All questions should be answered with complete, grammatically correct sentences.
- Sample computations should be labelled with their purpose and show all numbers with correct units and significant figures.
- Graphs should have axes numbered and labelled, a title, data points clearly marked (circles, crosses, etc.), and lines explained.
- Pages should be stapled or numbered with your name to avoid misplacement.

Experiment: Measurements and Density

This experiment demonstrates the proper use of measuring instruments, determination of density, and evaluation of precision and accuracy.

Objectives:

- To become familiar with mass and volume measurements.
- To determine the density of a metal bar and a salt solution using two different methods.
- To evaluate the precision and accuracy of the results.

Materials and Chemicals:

Cylindrical metal bars (Al, Cu, brass), measuring rules (graduated in mm), 20 or 25 mL transfer pipets, 50 mL beaker, graduated cylinders (10 mL and 50 mL or 100 mL), 125 mL Erlenmeyer flask, stopper, thermometers, balances with precision to 1 mg, saturated salt solutions ($NaCl$ and/or KCl).

Safety Precautions:

- Review the safety rules before starting.
- Take special care when inserting the bar into the graduated cylinder – do not drop it in; the glass cylinder may break.
- Pipetting should always be done using a suction device – never by mouth.

Part I: Measurements

A. Mass Measurements:

1. Zero the balance after cleaning the pan.
2. Measure the mass of a clean dry 50 mL beaker to the nearest 0.001 g.
3. Record the observation in ink in your lab notebook.
4. Remove the beaker, clean the pan, zero the balance, and re-weigh the same beaker.
5. Repeat step 4 one more time (total of three mass measurements).
6. Calculate the average mass of the beaker from the three measurements.
7. Repeat the weighing on a second balance (just one weighing).
8. Repeat on a third balance (just one weighing).

B. Volume Measurements:

Use of a pipet: Most volumetric pipets are designed to **deliver** rather than contain the specified volume. They are marked with 'TD'. Never pipet by mouth – always use a suction device.

1. Measure the temperature in the laboratory. Your teacher will provide the density of water at this temperature.
2. Use the same 50 mL beaker from Section A for determining the mass of each aliquot of water.
3. Measure 20 or 25 mL of water into the beaker using a pipet.
4. Record the volume of water to the appropriate number of significant figures.
5. Weigh the beaker and water to the nearest 0.001 g.
6. Calculate the mass of water in the beaker.
7. Use the mass and density of water to determine the volume of water measured.
8. Repeat steps 3–7 using a graduated cylinder instead of the pipet.
9. Repeat steps 3–7 using a 50 mL beaker to measure the water.

Part II: Density

A. Density of a Metal Bar:

1. Zero the balance. Weigh a metal bar to the nearest 0.001 g. Repeat the entire weighing operation twice.
2. Determine the volume of the metal bar by each of the following methods, making at least three measurements for each:

   Method I (Water Displacement): Insert the bar into a graduated cylinder with enough water to immerse it. Read the initial and final water levels (lowest point of the meniscus). Estimate the volume to one digit beyond the smallest scale division. Discard the water and repeat twice with different initial volumes.

   Method II (Dimensions): Measure the dimensions of the bar with a measuring stick ruled in centimetres. Repeat twice. Calculate the volume of the bar from these dimensions using the formula for a cylinder: $V = \pi r^2 h$.
3. For each method, calculate the average density and determine the relative error compared to the accepted value.

B. Density of a Salt Solution:

1. Weigh a 125 mL Erlenmeyer flask and stopper.
2. Using a 20.00 or 25.00 mL volumetric pipet, pipet the salt solution into the flask and reweigh. Repeat twice.
3. Calculate the mass of salt solution and the average density.
4. Weigh an empty, dry 10 mL graduated cylinder.
5. Fill with about 9–10 mL of salt solution, record the volume, and reweigh. Repeat twice.
6. Calculate the average density of the salt solution.
7. For each method, determine the relative average deviation. Which method is more precise?

Disposal:

Salt solutions: Do one of the following as indicated by your teacher:

- Recycle: Return the salt solution to its original container.

• Dilute 1:10 with tap water and flush down the sink with running water.

• Put in a waste bottle labeled 'inorganic waste'.

Unit Summary

• The **International System (SI)** uses seven base units: metre (m), kilogram (kg), second (s), ampere (A), kelvin (K), mole (mol), and candela (cd).

• **Derived units** (volume, density, speed, force, pressure, energy) are obtained by combining base units.

• **Temperature** is measured in Kelvin, Celsius, and Fahrenheit. Conversions: K = °C + 273.15, °F = (°C × 9/5) + 32, °C = (°F – 32) × 5/9.

• **SI prefixes** (kilo, milli, micro, nano, etc.) are used to scale units by powers of 10.

• All measurements have **uncertainty** due to systematic or random errors.

• **Accuracy** is closeness to the true value; **precision** is reproducibility of repeated measurements.

• **Significant figures** indicate the exactness of a measurement. Rules exist for counting sig figs and for calculations.

• **Scientific notation** (M × 10■) is used for very large or very small numbers.

• The **scientific method** is a systematic approach: observation → question → hypothesis → experiment → analysis → conclusion → communication.

• **Laboratory safety** is essential. Follow all rules, wear appropriate protection, and think before you act.

• A **laboratory report** should be neat, complete, and include all data with correct units and significant figures.

• **Density** is mass per unit volume (d = m/V). It is an intensive property used to identify substances.

• The **measurements and density experiment** demonstrates proper use of balances, pipets, graduated cylinders, and calculation of density.

Key Terms

Accuracy – closeness of a measured value to the true value.

Calibration – the process of adjusting an instrument to ensure accurate measurements.

Density – mass per unit volume (d = m/V).

Derived units – units obtained by combining SI base units.

Error – the difference between a measured value and the true value.

Exact numbers – numbers obtained from counting or definitions (considered to have infinite significant figures).

Precision – closeness of repeated measurements to each other.

SI units – the International System of Units (base units).

Scientific notation – expressing numbers as M × 10■.

Significant figures – meaningful digits in a measured or calculated quantity.

Uncertainty – the margin of error in a measurement.

End of Unit 2 – Measurements and Scientific Methods